CSCE 2501
Fundamentals of Database Systems

Project - Resale Properties

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Introduction

In this project, you will be building a database application for the real estate properties in Egypt that are offered for Resale, that allows users to get information about these properties in the country, the different developments, locations, etc.

The main site that provide such information is AqarExit website (https://aqarexit.com/en/opportunities?show=8000). However, the website has limited abilities for filtering or doing some higher level aggregations on the market size and state. Hence, your application will be used to provide more flexibility on that front.

Your task is to build a database system backend and an application that provide the resale market analytics and features described next.

Milestone I: Data Requirements

The database system you design should be based on the information provided on each property, its location, developer, type, and the other information provided on the property page (e.g. sample)

Your database design should store all the information about the resale property, including but not limited to: area, location, development, type (apartment, villa, etc.), number of bedrooms and bathrooms, list of amenities, description, payment method, down payment, gain over current value, etc. Many of the properties on the website are located in a development project, for such cases the development project needs to be recorded, including its name, location, developer, etc.

The system should also allow users to register on the system using their email addresses, and pick a username, gender, age, birthdate, and their main areas of focus. After registering, the users can add their reviews for any development project, providing their rating (1-10) and a textual review.

Milestone I: Database Design & Implementation

Tools: MySQL

Deadline: 3rd October, 2026

Description:
In this milestone, you are required to design the database ERD, create your database and schema in MySQL server for the data requirements presented on the previous slides.

Deliverables: (20 points)

- Entity-Relationship Diagram of the Real-Estate System
- Relational Model for your system
- A transcript for the SQL statements used to create your database and schema

Milestone II: Database Population

In this milestone, you are required to write a web crawler to crawl the pages from the AqarExit website for all the properties listed, parse the HTML you crawled and extract the relevant fields for populating the non-user tables in your schema. Note that you may need to extract some of the fields from the text fields in the web pages.

At the end of this milestone, you should have all your non-user tables populated with the extracted data. For the user tables, use sample test information to populate the relevant tables.

Make sure to rate-limit requests (Cloudflare is present; aggressive scraping risks IP blocks).

Milestone II: Web Crawling and Data Population

Tools: Scrappy, pycharm, BeautifulSoup, selenium, python (or any other crawler library / language)

Deadline: 31st October, 2026

Description:
In this milestone, you are required to implement your crawler, extract the data from the property finder website and populate your database

Deliverables: (50 points)

- Crawling script
- Populated MySQL database dump
- CSV files for the values in each table

Milestone III: Application Layer

In the final milestone, you are required to design a client application that is capable of connecting to the database hosted on remote MySQL server. The application should have the following functionalities:

- Register a user
- Add a new user review on a project
- View existing reviews of a project
- Show the top 10 projects with highest resale inventory, resale amount, and highest gain over market value by property type
- Show the location of a given development, along with the average price / sqm and the number of listings for each unit type
- Show all the properties of in a certain area, along with the average price / sqm for each unit type
- Show all the properties in a certain area in a given price range, with a given set of amenities
- Show the top 5 developers by the amount of resale listing they have, along with their avg price / sqm, avg. required down payment, and gain over market value
- Show all the properties by area, property type, and contract year
- Show the average installment price, avg down payment, number of properties, avg discount over market price, avg age of contract year, and avg market price by area and development project.

Milestone III: Application Layer

Tools: Any language / platform to implement your application (Web-based, GUI, command-line are all acceptable)

Deadline: 14th November, 2026

Description:
In this milestone, you are required to implement an application to perform various transactions / queries on your backend database system. The database has to be hosted on an actual server (you can use any of the free DB hosting services available (e.g. www.db4free.net)) and make sure your application is communicating with this service not the localhost

Deliverables: (30 points + 5 bonus points for GUI/Web-based apps)

- Application Implementation / Source Code + Executable
- Latest Dump of your database
- Demo